



Ministry of
IT & Telecom



Hands-on Exercise No. 3
DigiSkills 3.0 Batch-04
Artificial Intelligence Using Python

Total Marks: 10

Due Date: 24/09/2026

Instructions:

Please read the following instructions carefully before submitting this Hands-on Exercise:

- Use MS Word to prepare exercise solution.
- You may consult tutorials and videos if the concept is not clear.
- Your submitted exercise will not be considered/counted if:
 - It is submitted after due date.
 - It is not in the required format (.doc or .docx)
 - It does not open, or file is corrupt.
 - It is copied (partial or full) from any source (websites, forums, students, etc.)

Learning Outcome:

After completing this hands-on exercise, you shall be able to:

- **Load and explore a text dataset** using Pandas by examining records, dimensions, column names, data types, missing values, and class distribution.
- **Preprocess textual comments** by cleaning text, separating input text from the target variable, encoding labels, and splitting the data into training and testing sets.
- **Apply NLP preprocessing using tokenization and sequence padding** to convert text into a format suitable for Deep Learning.
- **Build, train and evaluate any Deep Learning Model (ANN, CNN, LSTM)** using TensorFlow/Keras for multi-class toxic comment classification.
- **Visualize model performance** using accuracy/loss curves and a confusion matrix and test the trained model on new comments.

Problem Statement

In this hands-on exercise, you will develop and evaluate a Deep Learning model for **Toxic Comment Classification** using the provided **toxic_comments_dataset.csv** dataset. The exercise will begin by loading and exploring the dataset using Pandas, including viewing sample records, checking its dimensions, column names, data types, missing values, and toxicity label distribution. You will then preprocess the comment text by cleaning the data, encoding the toxicity labels, and splitting the dataset into training and testing sets. After that, you will apply **Natural Language Processing (NLP) techniques such as text tokenization and sequence padding** to prepare the comments for Deep Learning.

Next, you will build and train any **Deep Learning model (ANN, CNN, LSTM)** using TensorFlow/Keras for classifying comments into their respective toxicity categories. Finally, you will evaluate the trained model using accuracy, loss, classification report, confusion matrix, and training/validation performance curves, and test the model on new comments to observe its predictions.

The purpose of this exercise is to provide practical experience with the complete NLP and Deep Learning workflow, from text preprocessing and tokenization to DL model development, training, prediction, and performance evaluation.

You can use **Jupyter Notebook, Google Colab, VS Code, or any other Python programming environment/IDE** to write and execute your code. After completing each task, take screenshots of your **code and its output** and paste them into a single **MS Word (.doc/.docx)** file for submission.

Tasks:

Download the “**toxic_comments_dataset.csv**” file from LMS with the following link:

https://lms.digiskills.pk/Courses/AIP301/Downloads/toxic_comments_dataset.csv

Task 1: Dataset Loading and Initial Exploration

(Marks: 2)

Load the **toxic_comments_dataset.csv** dataset using **Pandas** and display the following:

1. Display the first 10 and last 8 records and check the dataset shape.
2. Display the column names and data types of all columns.
3. Display the statistical summary of the numerical columns.

Task 2: Text Cleaning, Preprocessing and Data Quality, Target Analysis

(Marks: 2)

1. Convert comments to lowercase and handle missing values and duplicate comments.
2. Check the number of duplicate records and display the number of missing values in each column.
3. Clean the comment text by converting it to lowercase and removing tags, punctuation, numbers, special characters, extra spaces, emojis, and unnecessary whitespace.

Task 3: NLP Tokenization and Sequence Padding***(Marks: 2)***

1. Create a Keras Tokenizer using the training text only.
2. Convert training and testing comments into numerical sequences.
3. Select a suitable maximum sequence length and apply padding/truncation.

Task 4: Train and Validation the Deep Learning Model:***(Marks: 2)***

Using **TensorFlow/Keras**, build a multi-class Deep Learning model by performing the following tasks:

1. Including Embedding, Model, Dropout, Dense, and Softmax layers.
2. Train the DL model using the training data set using appropriate batch size.
3. Use a validation split and train the model for a suitable number of 10–20 epochs.

Task 5: Evaluate, Plotting and Visualize Model Performance***(Marks: 2)***

1. Create and display a confusion matrix for the test predictions.
2. Evaluate the trained model on the test dataset and display the test accuracy and loss.
3. Generate and display a classification report containing precision, recall, and F1-score for each class.

NOTE:

After completing all tasks, save the code file at your end. Just capture screenshots of your Python code and the corresponding outputs and paste them into a single Microsoft Word (.doc/.docx) document for submission.

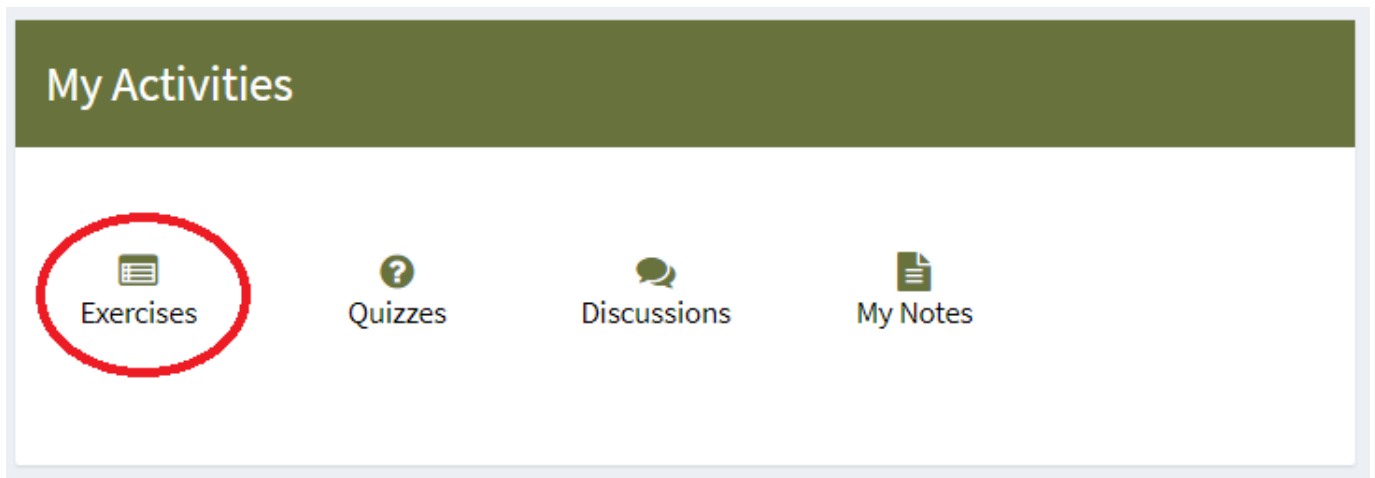
No need to send/attach code file; only provide screenshots of the above tasks.

BEST OF LUCK 😊

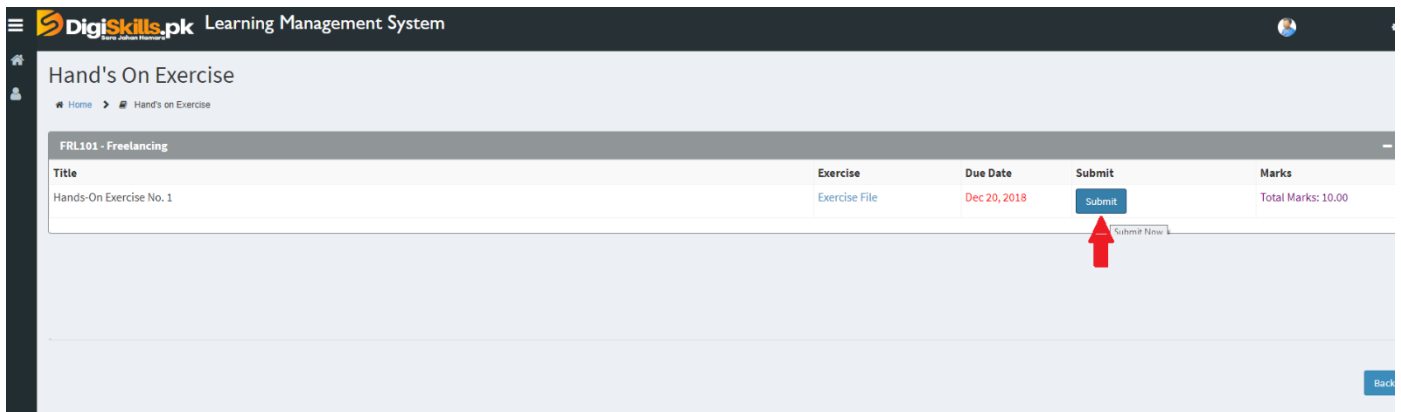
How to submit solution file on LMS?

Please perform the following steps for submitting your solution using LMS:

- 1) Login to the LMS
- 2) Click on the **Exercises** button within the **My Activities** section



- 3) Click on the submit button to upload your Solution.



- 4) Keep in mind to upload your Solution in .doc or .docx format